Google Cloud H2S

Gen AI Academy

APAC Edition

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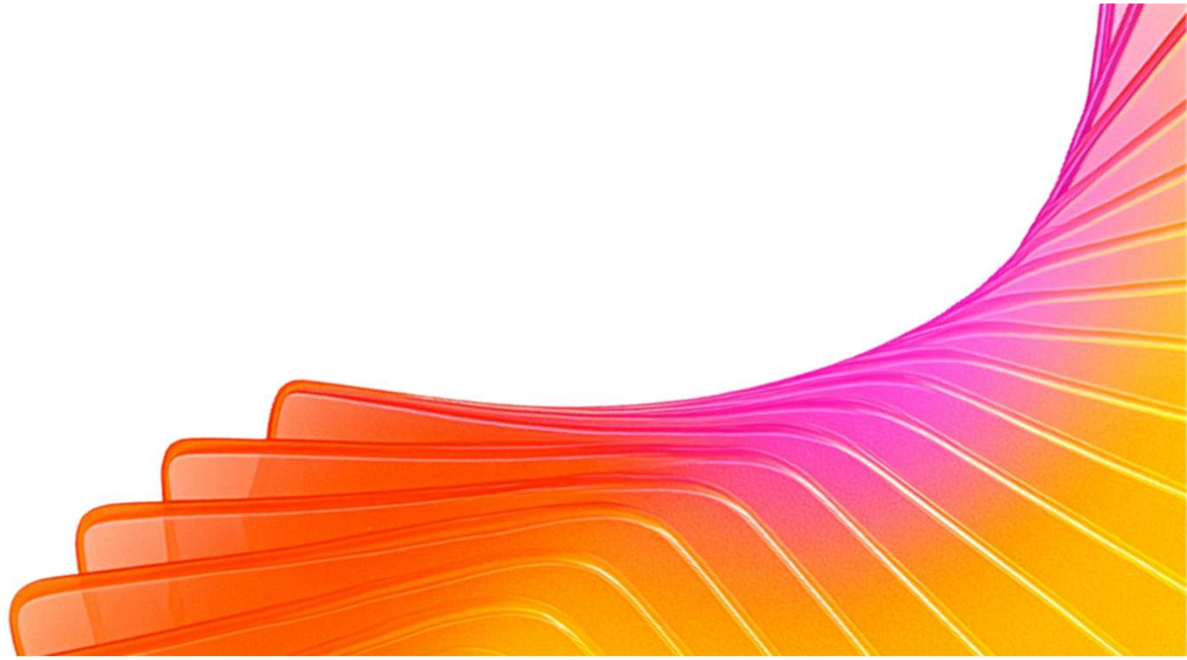

Participant Details

Participant Name: T S Renukesh

Problem Statement: AI-Powered Decision Intelligence Platform — turning community data into decisions (Civic & Governance)

Project: JanSetu AI — MPLADS Decision Copilot for Constituency Development

Brief about the idea

The problem: Rs 1,729 crore of MPLADS constituency-development funds lie unspent across India; only ~53% of released funds have been utilized since 2019, and 62% of MPs miss the mandatory 22.5% SC/ST-area allocation. Money and citizen need both exist — the missing piece is the DECISION between them.

Existing tools collect complaints (CPGRAMS, Swachhata) or track works after sanction (e-SAKSHI). None helps an MP's office decide WHICH works to sanction.

JanSetu AI ('bridge to the people') is that missing decision layer: citizens report civic issues by photo, voice (12 Indian languages) or text — even offline; AI fuses this demand signal with ward demographics, live spike detection and the MPLADS fund ledger to produce a ranked list of costed, compliant, sanctionable works — each with a one-click draft recommendation letter to the District Authority.

One deployment serves a constituency of ~25 lakh citizens; 543 constituencies nationwide.

Our approach — translating the problem into a working solution on Google Cloud

1. Understand: mapped the fund-utilization pipeline and found the gap — no decision support between citizen demand and fund sanction.

2. Ingest inclusively: multimodal citizen intake — Gemini Vision classifies photos into structured tickets (category, department, severity); voice input via on-device speech in 12 Indian languages; offline-first PWA queues reports till signal returns.

3. Analyze: Gemini NL-to-SQL over grievances + ward demographics + MPLADS ledger (guard-railed, read-only); statistical spike detection per ward x category; BigQuery mode for constituency-scale analytics.

4. Decide: Gemini synthesizes demand, spikes, demographics, fund balance and the 22.5% SC/ST mandate into a ranked Priority Works briefing with corrective amounts (e.g. 'sanction Rs 106 lakh in SC/ST wards to close the gap').

5. Act: one-click formal recommendation letter to the District Magistrate; multilingual responses via Cloud Translation API; deployed on Cloud Run (asia-south1).

Resilience: every AI call degrades gracefully - Gemini -> Groq (free) -> built-in Lite engine - with user-facing notices; the platform never breaks, even with zero API keys.

Opportunities — how it differs & the USP

How it differs from existing solutions:

CPGRAMS / IGMS 2.0: collects & clusters complaints — no fund or works linkage. Swachhata app: photo complaints only. e-SAKSHI: tracks works AFTER sanction. Commercial constituency CRMs: complaint management, no compliance or fund intelligence.

USP — the only platform that closes the loop from citizen voice to sanctioned work:

• Fuses demand signal + MPLADS fund ledger + SC/ST 22.5% compliance + ward demographics into ranked sanctionable works — genuinely new.

• Compliance radar with corrective amounts, not just alerts.

• One-click DA letter converts AI insight into official action.

• Inclusion as architecture: voice-first, 12 languages, photo reporting for non-readers, offline-first, no login, no PII.

• Always-functional: 3-tier AI degradation guarantees uptime within free tiers.

List of features offered by the solution

For citizens:

• Photo -> AI-classified grievance ticket (Gemini Vision) • Voice input in 12 Indian languages; answers read aloud (11 scripts) • Works offline - reports queue & auto-sync • Privacy: consent screen, no Aadhaar/name/address

For the MP's office:

• Natural-language Q&A over all constituency data, in any Indian language, with charts • Ward-level demand hotspot map (Google Maps) • Statistical spike alerts (explainable, auditable) • MPLADS fund tracker: unspent balance, SC/ST 22.5% compliance radar with corrective amounts • Priority Works briefing: ranked, costed, compliant, sanctionable works • One-click draft recommendation letter to District Authority • Office authentication + rate limiting

Platform:

• Graceful degradation: Gemini -> Groq -> Lite engine; Google Maps -> OSM; Cloud Translation -> Gemini -> Groq - with transparent provider notices • BigQuery mode for scale • PWA installable on any smartphone

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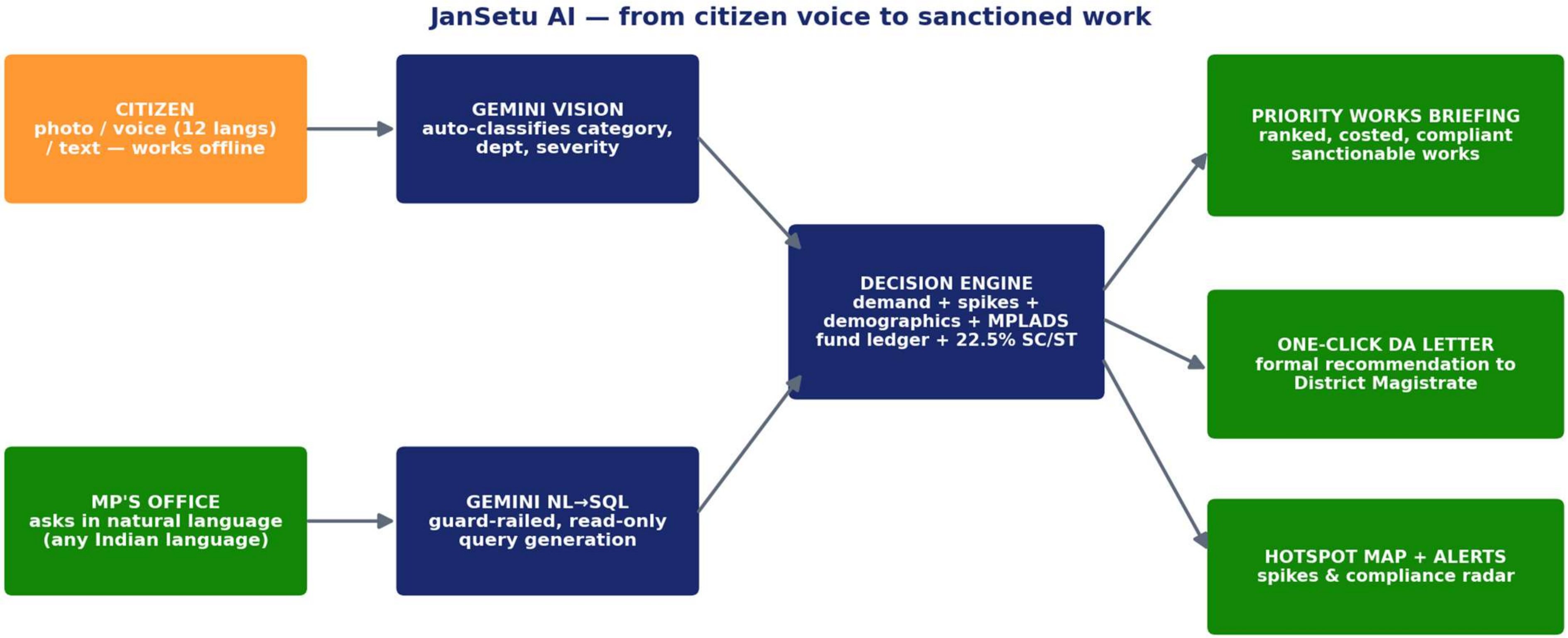
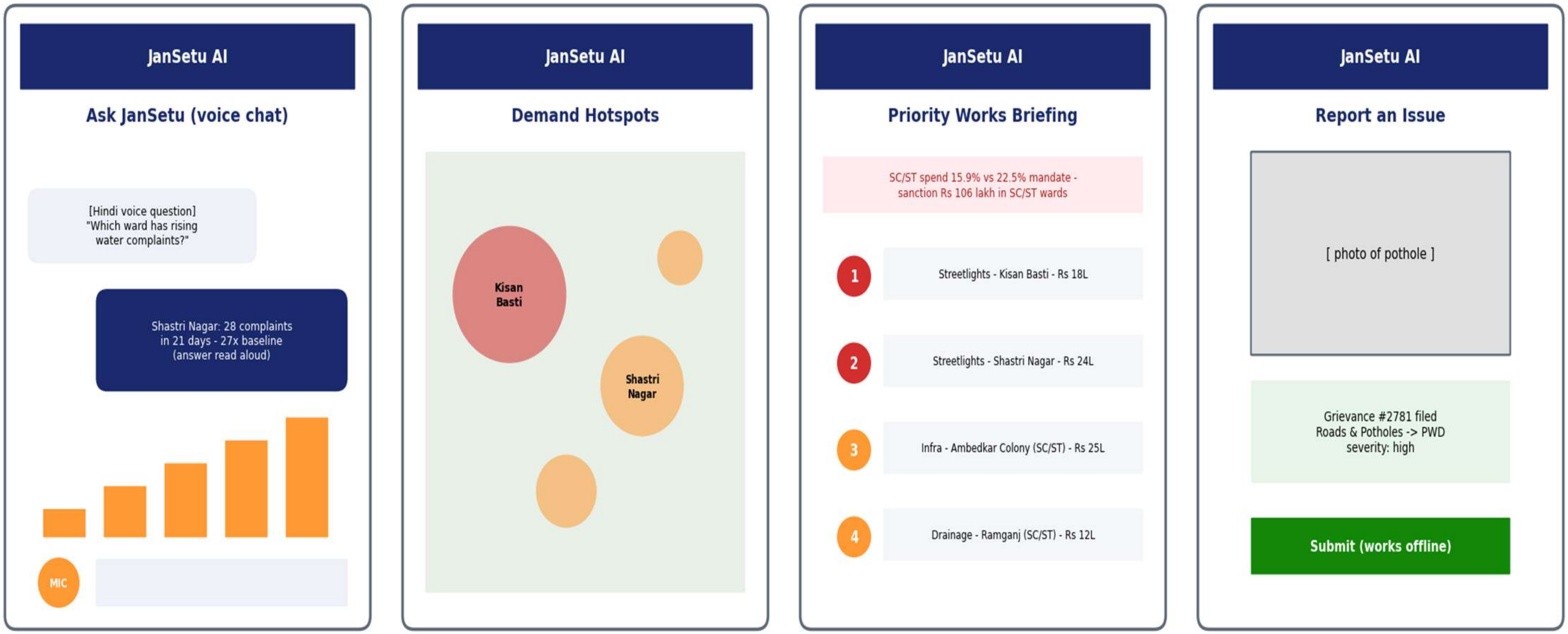
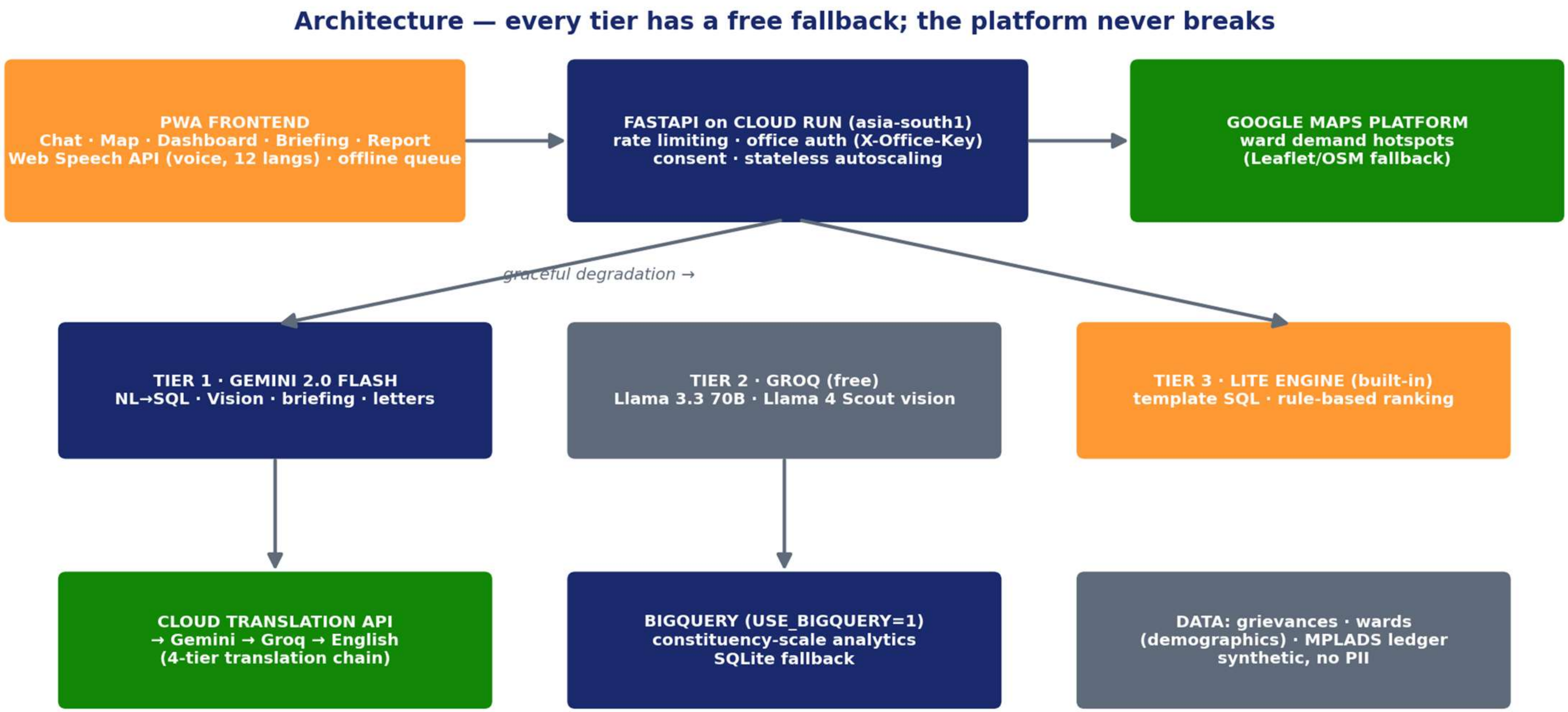
Process flow diagram or Use-case diagram



Wireframes/Mock diagrams of the proposed solution



Architecture diagram of the proposed solution:



Technologies / Google / Nvidia Services used in the solution

(Why did you choose your specific AI stack and system design, and how does it support scalability and real-world deployment?)

Google stack (5 services):

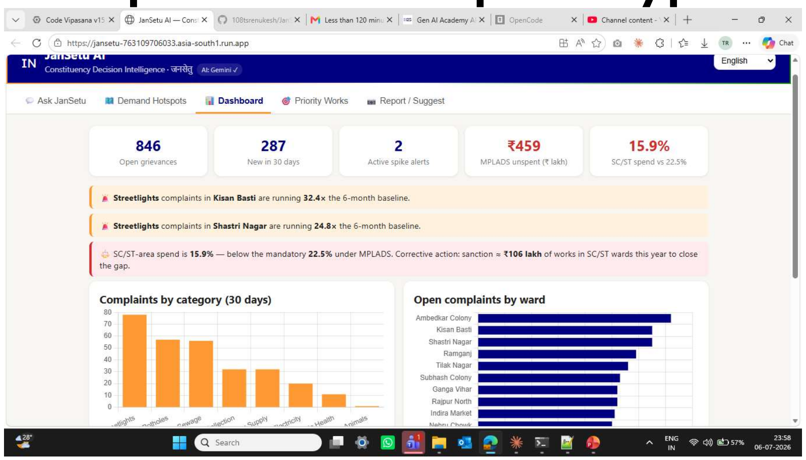
• Gemini 2.0 Flash (AI Studio) - NL->SQL, Vision triage, briefing synthesis, letter drafting, translation fallback • BigQuery - constituency-scale analytics (USE_BIGQUERY=1 switch, loader script) • Cloud Translation API - multilingual responses • Cloud Run (asia-south1) - stateless autoscaling deploy • Google Maps Platform - demand hotspots

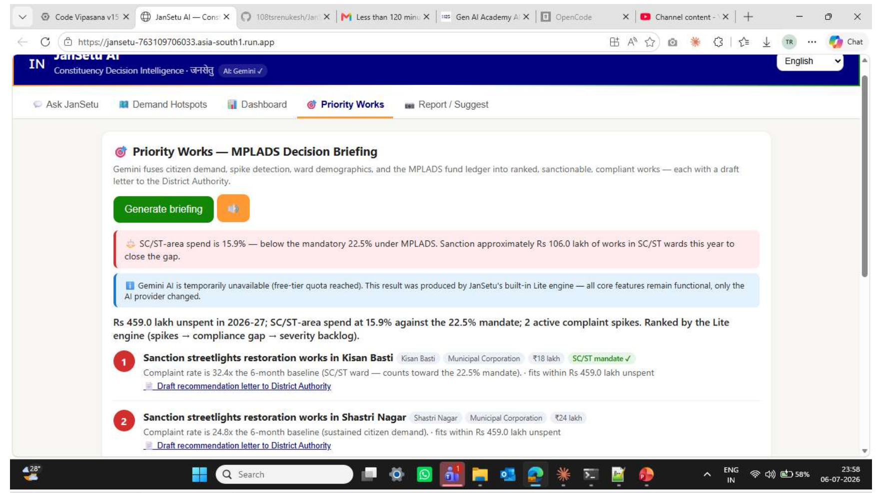
Supporting: FastAPI (Python), Groq free tier (fallback LLM/vision), Web Speech API (free on-device voice), Leaflet/OSM (map fallback), SQLite -> BigQuery data layer.

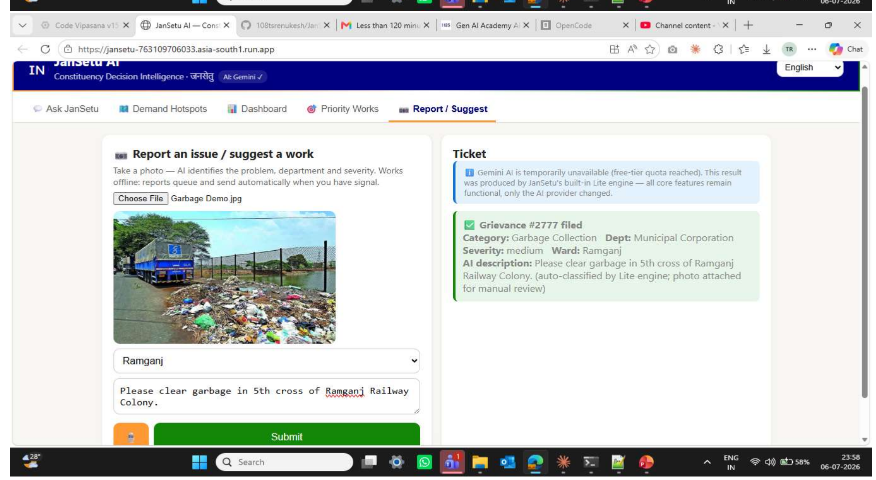
Why this design:

• Explainable statistics for spike detection (not black-box ML) - governance decisions must be auditable. • Read-only guard-railed SQL - the LLM can never mutate data. • Free-tier-first with graceful degradation - a real MP office can run this at near-zero cost; the demo cannot break. • Stateless Cloud Run + BigQuery = one constituency today, 543 tomorrow; new constituency onboarding under a day. • India data residency, DPDP-aligned consent, no PII - deployable in government context.

Snapshots of the prototype







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Thank you

JanSetu AI — from citizen voice to sanctioned work

GitHub: [108tsrenukesh/JanSetu-AI-MPLADS-Decision-Copilot]

Live demo: [JanSetu AI — Constituency Decision Intelligence]

Demo video:

Rs 1,729 crore is waiting. The decisions just got easier.